“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 12

1

Any Reminders? Any Questions?

- I have regular office hours Mondays via Webex 1-2pm
- I have regular office hours Thursdays in-person 3:30-4:30pm
- Calc 3 Calc Night: MONT 104 at 6:30-8:30pm on Thursdays!

2

ALVARO: Start the recording!

3

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Double Integrals in Polar

4

Today – Double Integrals in Polar Coordinates!

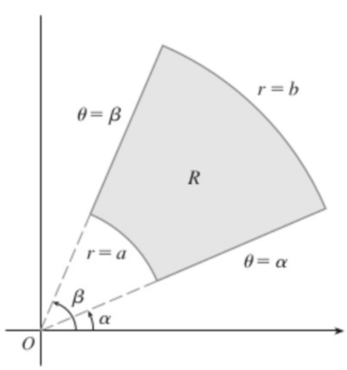
- Polar Coordinates
- Double Integrals in Polar Coordinates
- Change from Cartesian to Polar

5

Double Integrals in Polar Coordinates

If f is continuous on a polar rectangle R given by $0 \leqslant a \leqslant r \leqslant b$, $\alpha \leqslant \theta \leqslant \beta$, where $0 \leqslant \beta - \alpha \leqslant 2\pi$, then

$$\iint_R f(x, y)\, dA = \int_\alpha^\beta \int_a^b f(r\cos\theta, r\sin\theta)\, r\, dr\, d\theta$$



6

Example: Evaluate the value of the double integral

$$\iint_R (x + x^2 + y^2)dA$$

where R is the first quadrant of a circle of radius 2.

7

[Extra]

Example: Evaluate the value of the double integral

$$\iint_R (x + x^2 + y^2)dA$$

where R is the first quadrant of a circle of radius 2.

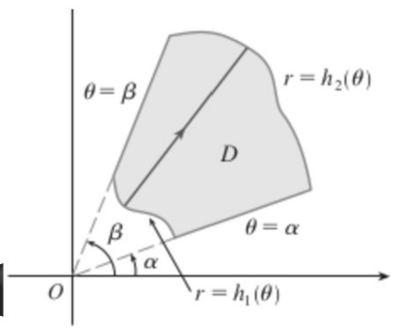
$$\sin^2(\theta) = \frac{1 - \cos(2\theta)}{2}$$

8

Integrals in Polar Coordinates of Type I

$$D = \{(r, \theta) \mid \alpha \leqslant \theta \leqslant \beta, h_1(\theta) \leqslant r \leqslant h_2(\theta)\}$$

$\boxed{3}$ If f is continuous on a polar region of the form



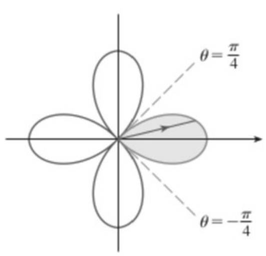
$$D = \{(r, \theta) \mid \alpha \leqslant \theta \leqslant \beta, h_1(\theta) \leqslant r \leqslant h_2(\theta)\}$$

$$\iint_D f(x, y)\, dA = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} f(r\cos\theta, r\sin\theta)\, r\, dr\, d\theta$$

9

Example: Find the area enclosed by one loop of the four-leaved rose

$$r = \cos(2\theta)$$



10

[Extra]

Example: Find the area enclosed by one loop of the four-leaved rose

$$r = \cos(2\theta)$$

$$\cos^2(\theta) = \frac{1 + \cos(2\theta)}{2}$$

11

Questions?

12

ALVARO: Start the recording!

13

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Triple Integrals

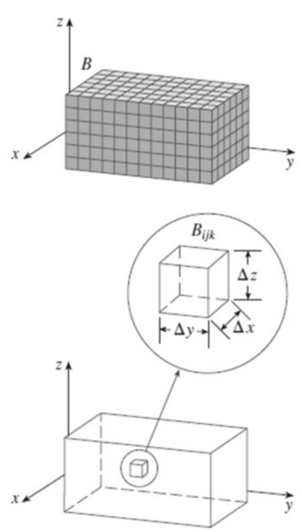
14

Today – Double Integrals in Polar Coordinates!

- Triple Integrals in Boxes
- Triple Integrals in Regions
- Changing the Order of Integration

15

Riemann Integral in 3 Variables



$$B = \{(x, y, z) \mid a \leqslant x \leqslant b, c \leqslant y \leqslant d, r \leqslant z \leqslant s\}$$

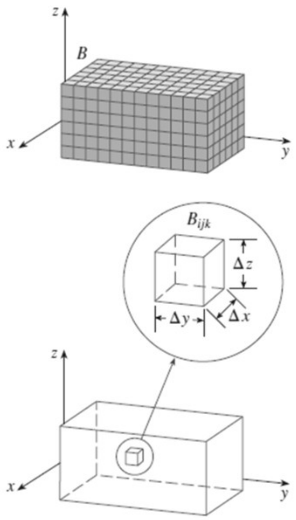
The **triple integral** of f over the box B is

$$\iiint_B f(x, y, z)\ dV = \lim_{l,\, m,\, n \to \infty} \sum_{i=1}^{l} \sum_{j=1}^{m} \sum_{k=1}^{n} f\left(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*\right)\ \Delta V$$

if this limit exists.

16

Riemann Integral in 3 Variables



$$B = \{(x, y, z) \mid a \leqslant x \leqslant b, c \leqslant y \leqslant d, r \leqslant z \leqslant s\}$$

4 Fubini's Theorem for Triple Integrals

If f is continuous on the rectangular box $B = [a, b] \times [c, d] \times [r, s]$, then

$$\iiint\limits_B f(x, y, z)\, dV = \int_r^s \int_c^d \int_a^b f(x, y, z)\, dx\, dy\, dz$$

17

Example: Evaluate the value of the triple integral

$$\iiint_B \; xyz^2 \, dV$$

where B is the box $[0,1] \times [-1,2] \times [0,3]$.

18

[Extra]

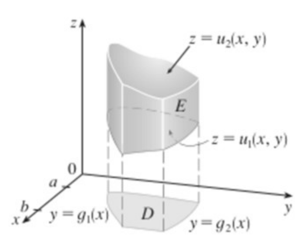
Example: Evaluate the value of the triple integral

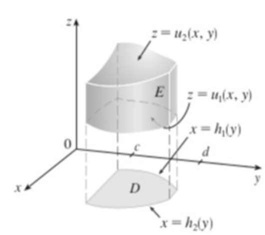
$$\iiint_B \; xyz^2 \, dV$$

where B is the box $[0,1] \times [-1,2] \times [0,3]$.

19

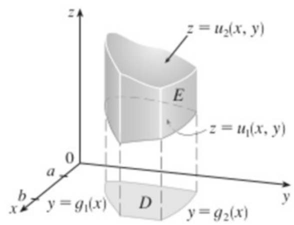
Triple Integrals in General Regions – Define the Region

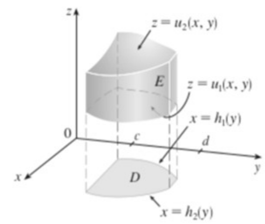




20

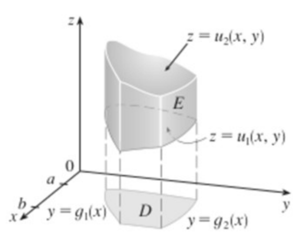
Triple Integrals in General Regions – Set up the Integral

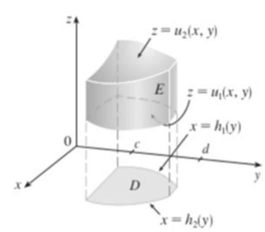




21

Triple Integrals in General Regions





$$\iiint_E f(x, y, z)\, dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{u_1(x,\, y)}^{u_2(x,\, y)} f(x, y, z)\, dz\, dx\, dy$$

$$\iiint_E f(x, y, z)\, dV = \int_c^d \int_{h_1(y)}^{h_2(y)} \int_{u_1(x,\, y)}^{u_2(x,\, y)} f(x, y, z)\, dz\, dx\, dy$$

22

Example: Evaluate the value of the triple integral

$$\iiint_E z\, dV$$

where E is the solid in the first octant bounded by the surface $z = 12xy$ and the planes $y = x$ and $x = 1$.

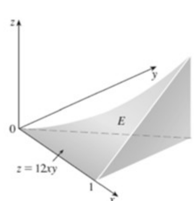
23

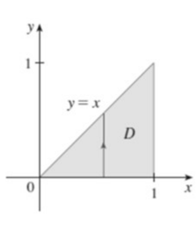
[Extra]

Example: Evaluate the value of the triple integral

$$\iiint_E z\, dV$$

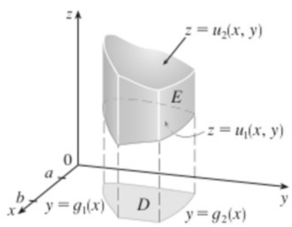
where E is the solid in the first octant bounded by the surface $z = 12xy$ and the planes $y = x$ and $x = 1$.

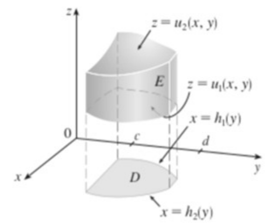




24

Triple Integrals in General Regions: Type I, …



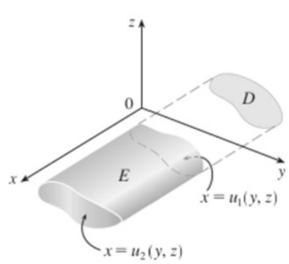


$$\iiint_E f(x, y, z)\, dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{u_1(x,\, y)}^{u_2(x,\, y)} f(x, y, z)\, dz\, dx\, dy$$

$$\iiint_E f(x, y, z)\, dV = \int_c^d \int_{h_1(y)}^{h_2(y)} \int_{u_1(x,\, y)}^{u_2(x,\, y)} f(x, y, z)\, dz\, dx\, dy$$

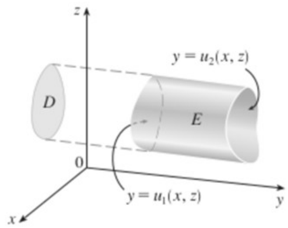
25

Triple Integrals in General Regions: Type II



26

Triple Integrals in General Regions: Type III



27

Example: Evaluate the value of the triple integral

$$\iiint_E \sqrt{x^2+y^2}\, dV$$

where E is the solid bounded by the surface $y = x^2 + z^2$ and the plane $y = 4$.

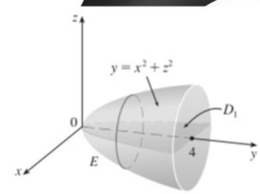
28

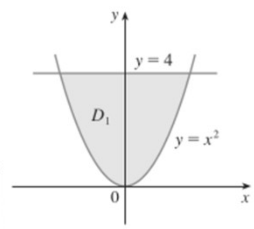
[Extra]

Example: Evaluate the value of the triple integral

$$\iiint_E \sqrt{x^2+y^2}\, dV$$

where E is the solid bounded by the surface $y = x^2 + z^2$ and the plane $y = 4$.





29

Changing the Order of Integration

Example: Rewrite the following triple integral as a Type I, Type II, and Type III integrals

$$\int_0^1 \int_0^{x^2} \int_0^{y} (x+y+z)\, dz\, dy\, dx$$

30

Changing the Order of Integration

Example: Rewrite the following triple integral as a Type I, Type II, and Type III integrals

$$\int_0^1 \int_0^{x^2} \int_0^{y} (x+y+z)\, dz\, dy\, dx$$

31

Changing the Order of Integration

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32

Changing the Order of Integration

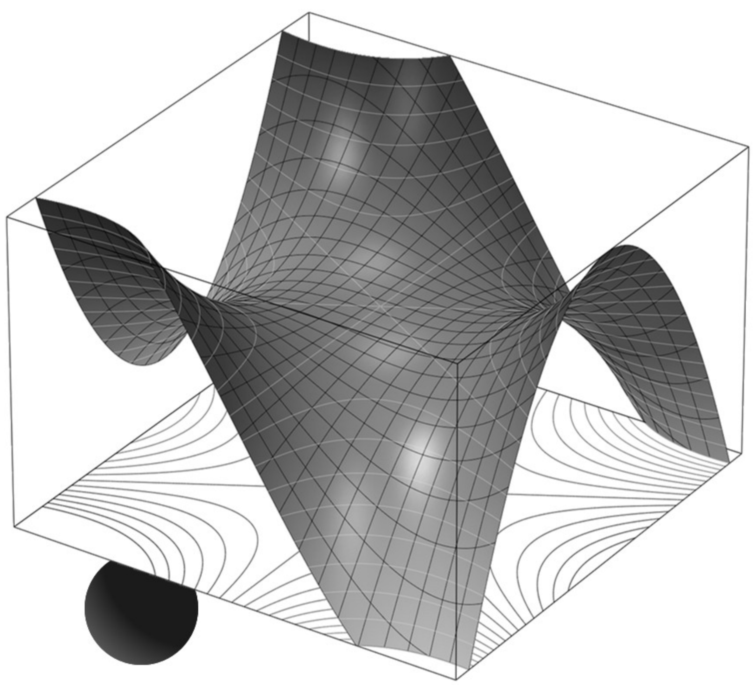
Example: Rewrite the following triple integral as a Type I, Type II, and Type III integrals

$$\int_0^1 \int_0^{x^2} \int_0^{y} (x + y + z)\, dz\, dy\, dx$$

33

Thank you

Until next time.



34